Adaptive Time-Stepping for Variational QITE: An RKF-Type Controller with a Hardware-Noise Floor

1 Setup: QITE as an ODE in parameter space

The variational QITE update solves, at each imaginary time τ,

$$M(\boldsymbol{\theta})\,\dot{\boldsymbol{\theta}} = Y(\boldsymbol{\theta}), \qquad \boldsymbol{\theta}(\tau+\Delta\tau) \approx \boldsymbol{\theta}(\tau) + \dot{\boldsymbol{\theta}}\,\Delta\tau, \tag{1}$$

i.e. an *explicit Euler* discretization of the autonomous flow

$$\dot{\boldsymbol{\theta}} = f(\boldsymbol{\theta}) := M^{+}(\boldsymbol{\theta})\,Y(\boldsymbol{\theta}), \tag{2}$$

where M^{+} denotes the (regularized) pseudoinverse. Since (2) is just an ODE whose right-hand side happens to be evaluated by a quantum processor, any classical adaptive integrator applies *formally*. The Runge–Kutta–Fehlberg (RKF) idea is to advance with an embedded pair of orders $(p, p+1)$,

$$\mathbf{k}_i = f\Big(\boldsymbol{\theta}_n + \Delta\tau \sum_{j<i} a_{ij}\,\mathbf{k}_j\Big), \qquad i = 1,\dots,s, \tag{3}$$

$$\boldsymbol{\theta}^{(p)}_{n+1} = \boldsymbol{\theta}_n + \Delta\tau \sum_i b_i \mathbf{k}_i, \qquad \boldsymbol{\theta}^{(p+1)}_{n+1} = \boldsymbol{\theta}_n + \Delta\tau \sum_i b^{*}_i \mathbf{k}_i, \tag{4}$$

and to use the embedded difference as a local truncation-error estimate,

$$e_n := \Big\|\boldsymbol{\theta}^{(p+1)}_{n+1} - \boldsymbol{\theta}^{(p)}_{n+1}\Big\| = \Delta\tau\,\|\textstyle\sum_i (b^{*}_i - b_i)\,\mathbf{k}_i\|, \qquad \Delta\tau_{\text{new}} = \Delta\tau\,\left(\frac{\text{tol}}{e_n}\right)^{1/(p+1)}. \tag{5}$$

Two features of the quantum setting break the classical analysis: (i) each stage $\mathbf{k}_i$ requires a full round of Hadamard-test circuits to estimate M and Y at a shifted parameter point — stages are *expensive*, scaling as $O(N^2)$ circuits per stage; and (ii) each $\mathbf{k}_i$ is corrupted, $\tilde{\mathbf{k}}_i = \mathbf{k}_i + \Delta\mathbf{k}_i$, with $\|\Delta\mathbf{k}_i\|/\|\mathbf{k}_i\|$ bounded by the hardware-noise results below. The error estimator e_n therefore measures truncation error *plus* noise, and the controller (5) must be modified accordingly.

2 The hardware-noise floor on the error estimator

Let ε_{hw} denote the relative error in a single evaluation of f. For probabilistic Markovian gate noise of strength p acting after each of the N gates, one has the general bound

$$\frac{\big\|\Delta\dot{\boldsymbol{\theta}}\big\|}{\big\|\dot{\boldsymbol{\theta}}\big\|} \;\le\; \operatorname{cond}(M)\,\Big(\big(1-(1-p)^{2N}\big)\Big(1+\tfrac{N}{2\|M\|}\Big) + \big(1-(1-p)^{N}\big)\Big(1+\tfrac{\sqrt{N}}{\sqrt{2}\|Y\|}\Big)\Big) =: B(p,N), \tag{6}$$

and for partially depolarizing noise the *exact* relation

$$\tilde{f} = (1-p)^{-N} f \qquad \Longrightarrow \qquad \frac{\left\|\Delta\dot{\boldsymbol{\theta}}\right\|}{\left\|\dot{\boldsymbol{\theta}}\right\|} = \frac{1-(1-p)^N}{(1-p)^N}. \tag{7}$$

Equation (7) is a pure *bias* and should be removed by rescaling $\tilde{f} \mapsto (1-p)^N \tilde{f}$ *before* the integrator ever sees the data; what remains is a residual stochastic error $\varepsilon_{\rm res}$ (shot noise, non-depolarizing components, imperfect knowledge of p), bounded in the worst case by (6).

The estimator in (5) then satisfies, schematically,

$$e_n^2 \;\approx\; \underbrace{C_{\rm tr}^2\,\Delta\tau^{2(p+1)}}_{\text{truncation}} \;+\; \underbrace{\Delta\tau^2\,\kappa_b^2\,\varepsilon_{\rm res}^2\left\|\dot{\boldsymbol{\theta}}\right\|^2}_{\text{noise floor}}, \qquad \kappa_b := \Big(\textstyle\sum_i (b_i^* - b_i)^2\Big)^{1/2}, \tag{8}$$

assuming approximately independent stage errors. The crucial observation: the noise term scales as $\Delta\tau^1$ while truncation scales as $\Delta\tau^{p+1}$. Once

$$\Delta\tau \;\lesssim\; \Delta\tau_{\rm floor} := \left(\frac{\kappa_b\,\varepsilon_{\rm res}\left\|\dot{\boldsymbol{\theta}}\right\|}{C_{\rm tr}}\right)^{1/p}, \tag{9}$$

the estimator returns pure noise, the controller can no longer distinguish step-size reduction from statistical fluctuation, and naive RKF will either stall ($\Delta\tau \to 0$) or oscillate. **Any adaptive QITE scheme must therefore enforce tol $\geq$ (noise floor).**

3 Attack routes

3.1 Route 1: Low-order embedded pairs with FSAL

Classical RKF45 uses $s = 6$ stages; each stage costs $O(N^2)$ circuits, and — decisively — hardware noise caps the achievable per-step accuracy at $\varepsilon_{\rm res}\left\|\dot{\boldsymbol{\theta}}\right\|\Delta\tau$ regardless of integrator order. High order buys nothing below the floor. The natural choices are:

- **Heun–Euler 2(1)**: $s = 2$, error estimate $e_n = \frac{\Delta\tau}{2}\|\mathbf{k}_2 - \mathbf{k}_1\|$. Doubles the quantum cost per step relative to Euler but permits much larger accepted steps.
- **Bogacki–Shampine 3(2)**: $s = 4$ with the *first-same-as-last* (FSAL) property — the final stage of an accepted step is reused as $\mathbf{k}_1$ of the next, so the marginal cost is 3 evaluations per step.

A useful accounting metric is quantum cost per unit imaginary time,

$$\mathcal{C}(\Delta\tau) = \frac{s_{\rm eff}\cdot n_{\rm circ}(N)\cdot n_{\rm shots}}{\Delta\tau}, \tag{10}$$

to be minimized subject to $e_n \leq$ tol. Since the noise floor forces tol $= \Theta(\Delta\tau\,\varepsilon_{\rm res}\left\|\dot{\boldsymbol{\theta}}\right\|)$, the optimum sits at the *largest* $\Delta\tau$ for which truncation error just reaches the floor:

$$\boxed{\Delta\tau^\star \simeq \left(\frac{\varepsilon_{\rm res}\left\|\dot{\boldsymbol{\theta}}\right\|\kappa_b}{C_{\rm tr}}\right)^{1/p}} \tag{11}$$

— the adaptive controller, given a noise-aware tolerance, finds this automatically. Note the inversion of classical logic: on NISQ hardware, adaptivity is valuable not to make steps small where the flow is fast, but to make them *as large as noise permits* where the flow is slow (e.g. near convergence, where $\|Y\| \to 0$ and $\dot{\boldsymbol{\theta}}$ flattens).

3.2 Route 2: Step-doubling (Richardson) instead of embedded pairs

Advance once with step $\Delta\tau$ and twice with $\Delta\tau/2$ using plain Euler:

$$e_n = \frac{\left\| \boldsymbol{\theta}^{(\Delta\tau)} - \boldsymbol{\theta}^{(\Delta\tau/2,\times 2)} \right\|}{2^p - 1}, \qquad p = 1. \tag{12}$$

Cost: 3 evaluations of f per adapted step, but implementable with zero change to the existing QITE measurement pipeline, and the extrapolated combination $2\boldsymbol{\theta}^{(\Delta\tau/2,\times 2)} - \boldsymbol{\theta}^{(\Delta\tau)}$ gives a second-order update for free. Because the two half-steps and the full step share the initial point, stage-noise correlations partially cancel in e_n, which slightly lowers the estimator's noise floor relative to (8).

3.3 Route 3: Cheap physics-based error indicators (no extra stages)

Embedded stages are the expensive part; QITE offers *free* monitors that can drive the controller instead:

1. **McLachlan residual.** The quantity $r_n := \left\| M\dot{\boldsymbol{\theta}} - Y \right\|$ (computable classically from the already-measured $M, Y, \dot{\boldsymbol{\theta}}$) measures how well the ansatz manifold tracks the exact flow; a spike signals that the linear model of the step is failing and $\Delta\tau$ should shrink.

2. **Energy monotonicity and the variance identity.** Exact imaginary-time flow obeys

$$\frac{dE}{d\tau} = -2\,\mathrm{Var}_\rho(H) \;\le\; 0. \tag{13}$$

   Measuring E and $\mathrm{Var}(H)$ at τ_n (one extra observable, not $O(N^2)$ circuits) yields a first-order *predicted* energy change $\widehat{\Delta E} = -2\,\mathrm{Var}(H)\,\Delta\tau$. Accept the step iff the measured ΔE agrees within a confidence band; reject/shrink on violation of monotonicity beyond shot noise. This is an embedded estimator in the *energy* rather than the parameters, and it is exactly the quantity one ultimately cares about for ground-state preparation.

3. **Local stiffness proxy.** The step-size stability limit of explicit Euler is set by the Jacobian spectrum of f; a cheap proxy is $\mathrm{cond}(M)$ (already needed for the error bound (6), and estimable without inverting M). Growing $\mathrm{cond}(M)$ flags stiffness $\Rightarrow$ cap $\Delta\tau$ or switch regularization (Route 5).

3.4 Route 4: Statistically-gated step control

Because e_n is a random variable, the accept/reject decision should be a hypothesis test, not a threshold comparison. Let σ_e be the propagated standard error of e_n (from shot-noise covariances of M, Y through M^+; a delta-method estimate suffices). Then:

$$\text{if } e_n - z\,\sigma_e > \text{tol}: \quad \text{reject, shrink via (5);} \tag{14}$$

$$\text{if } e_n + z\,\sigma_e < \text{tol}: \quad \text{accept, grow;} \tag{15}$$

$$\text{otherwise}: \quad \text{accept, hold } \Delta\tau \text{ (estimator is noise-dominated).} \tag{16}$$

Combine with a PI (Gustafsson-type) controller,

$$\Delta\tau_{n+1} = \Delta\tau_n \left(\frac{\mathrm{tol}}{e_n}\right)^{k_I} \left(\frac{e_{n-1}}{e_n}\right)^{k_P}, \qquad k_I \approx 0.3/(p{+}1),\ k_P \approx 0.4/(p{+}1), \tag{17}$$

which damps the step-size oscillations that a noisy estimator otherwise induces in the standard deadbeat controller.

3.5 Route 5: Stiffness-aware variants

Imaginary-time flow is gradient-like; near convergence the Jacobian of f has widely separated negative eigenvalues (ratio $\sim$ spectral gap structure of H projected on the ansatz tangent space), i.e. the ODE is stiff precisely where one spends most iterations. Options:

- **Linearly-implicit (Rosenbrock) steps**: treat the dominant linear part of f implicitly using a classically-stored approximation to the Jacobian (e.g. finite differences of measured $\dot{\boldsymbol{\theta}}$ across recent steps, or a quantum-geometric-tensor–based surrogate); no extra quantum cost beyond what adaptive stepping already measures.
- **Exponential/analytic substeps**: freeze M, Y over a macro-step and integrate the frozen linear system exactly, with the adaptive controller governing only the macro-step over which freezing is valid.
- **Regularization coupling**: if M^+ is Tikhonov-regularized with parameter λ, the truncation constant C_{tr} and the noise amplification $\mathrm{cond}(M_\lambda)$ trade off; λ and $\Delta\tau$ should be co-adapted, e.g. by minimizing the per-step total of (8) over both.

4 A concrete algorithm

Algorithm 1 Noise-aware adaptive QITE (Heun–Euler 2(1) variant)

Input: $\mathrm{tol}_{\mathrm{user}}$, noise level p, z, initial $\Delta\tau$
Measure $\tilde{M}, \tilde{Y}$; de-bias by (7); solve for $\mathbf{k}_1$
while $\tau < \tau_{\max}$ **do**
  Measure f at $\boldsymbol{\theta}_n + \Delta\tau\,\mathbf{k}_1 \to \mathbf{k}_2$ (de-biased)
  $e_n \leftarrow \frac{\Delta\tau}{2}\|\mathbf{k}_2 - \mathbf{k}_1\|$; estimate σ_e from shot statistics
  $\mathrm{tol} \leftarrow \max\{\mathrm{tol}_{\mathrm{user}},\ c\,\Delta\tau\,\hat{\varepsilon}_{\mathrm{res}}\|\mathbf{k}_1\|\}$ ▷ noise floor via (6) or measured
  Accept/reject via the statistical gate (Route 4)
  if accepted **then**
    $\boldsymbol{\theta}_{n+1} \leftarrow \boldsymbol{\theta}_n + \frac{\Delta\tau}{2}(\mathbf{k}_1 + \mathbf{k}_2)$; check ΔE against (13)
    $\mathbf{k}_1 \leftarrow \mathbf{k}_2$ at new point (re-measure; or FSAL if using BS3(2))
  end if
  Update $\Delta\tau$ with PI controller
end while

5 Discussion

Three structural conclusions. First, adaptivity and error *mitigation* interlock: the exact depolarizing rescaling (7) converts the dominant noise channel from a floor-setting error into a removable

bias, directly enlarging the regime where the RKF estimator is informative. Second, the noise floor argument shows the accumulated error over a horizon T decomposes as $\mathcal{E}_{\text{tot}} \approx C\,T\,\Delta\tau^p + T\,\varepsilon_{\text{res}}\left\|\dot{\boldsymbol{\theta}}\right\|$: the noise contribution is $\Delta\tau$-independent, so shrinking steps below $\Delta\tau^\star$ only multiplies measurement cost without improving accuracy — the controller's real job on NISQ hardware is to *maximize* the step. Third, the physics-based indicators (McLachlan residual, variance identity) provide error control at $O(1)$ extra circuits rather than $O(N^2)$ per stage, and are likely the practical entry point before full embedded pairs are attempted.